

Human-Supervised CFAR Autotuning for Maritime Sensing

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AIAA Region II Student Conference 2026

Motivation: CFAR Tuning Challenges

Problem: CFAR detectors require scene-specific parameter tuning

Real-world challenges:

- Heavy-tailed clutter (K-distribution, Pareto models)
- Terrain transitions and multipath interference
- Spatially nonhomogeneous background statistics
- No universal parameter set works across environments

Current practice: Manual trial-and-error

- Slow, subjective, not reproducible
- Operators adjust until display “looks right”
- No quantitative optimization objective

Our approach: Formalize operator knowledge as ground truth and optimize systematically

Background: CFAR Detection Variants

All variants use 2D sliding window with **guard cells** N_g and **reference cells** N_r

Threshold multiplier (exponential clutter):

$$\alpha = N \left(P_{\text{FA}}^{-1/N} - 1 \right)$$

CA-CFAR

- Mean of all reference cells
- Optimal for homogeneous clutter

$$Z^{\text{CA}} = \frac{1}{N} (S_{\text{outer}} - S_{\text{inner}})$$

Detection rule:

GO-CFAR

- Greatest of two half-windows
- Suppresses false alarms at edges

$$Z^{\text{GO}} = \max(\bar{Z}_1, \bar{Z}_2)$$

SO-CFAR

- Smallest of two half-windows
- Maintains sensitivity near edges

$$Z^{\text{SO}} = \min(\bar{Z}_1, \bar{Z}_2)$$

$$D_t(i, j) = \mathbb{K}[I_t(i, j) > \alpha \cdot Z_{i,j}]$$

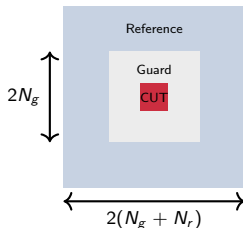
Problem Formulation

Goal: Find optimal CFAR variant $v \in \{\text{CA}, \text{GO}, \text{SO}\}$ and parameters $\theta = (N_g, N_r, P_{\text{FA}})$

Key insight: Integral images

- Pre-compute summed area table
- $O(1)$ query for any rectangle
- Enables fast noise map computation

2D Sliding Window:



Optimization challenge:

- Parameter space: $N_g \in [1, 10]$, $N_r \in [5, 30]$, $P_{\text{FA}} \in [10^{-6}, 10^{-2}]$
- Noise map depends on (N_g, N_r) only
- **Loop hoisting:** compute once per (N_g, N_r) , reuse across all P_{FA}

Speedup factor: Number of P_{FA} values tested

Annotation and Scoring Function

Blob tracing annotation:

- Operator clicks on detected target
- 8-connected flood fill captures exact shape
- Ground truth blob $\mathcal{B}_k = \text{pixel set}$

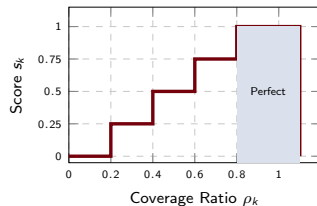
Coverage ratio:

$$\rho_k = \frac{\sum_{p \in \mathcal{B}_k} D_t(p)}{|\mathcal{B}_k|}$$

Score mapping:

$$s_k = \begin{cases} 1.00 & \text{if } 0.8 \leq \rho_k \leq 1.1 \\ 0.75 & \text{if } 0.6 \leq \rho_k < 0.8 \\ 0.50 & \text{if } 0.4 \leq \rho_k < 0.6 \\ 0.25 & \text{if } 0.2 \leq \rho_k < 0.4 \\ 0.00 & \text{if } \rho_k < 0.2 \end{cases}$$

Scoring function visualization:



Frame score (weighted):

$$S_{\text{frame}} = 0.7 \cdot \min_k s_k + 0.3 \cdot \frac{1}{K} \sum_{k=1}^K s_k$$

Heavy penalty for missing any target

Grid Search with Loop Hoisting

Algorithm 1 Optimized Grid Search

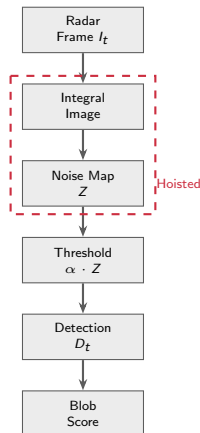
```
1: Input: frames, blobs, variant
2: Pre-compute integral images
3: for each  $(N_g, N_r)$  do
4:   Compute noise map  $Z \leftarrow$  expensive
5:   for each  $P_{FA}$  do
6:      $\alpha \leftarrow N(P_{FA}^{-1/N} - 1)$ 
7:      $D_t \leftarrow \mathbb{K}[I_t > \alpha Z]$ 
8:     Score detections vs. blobs
9:   end for
10: end for
11: Return sorted results
```

Temporal stability:

$$\text{Stability} = \frac{1}{K} \sum_{k=1}^K \frac{c_k}{T}$$

where c_k = frames detecting target k

Processing pipeline:



Performance: ~ 500 configs in < 10 seconds

System Architecture: CFAR Studio

Fully client-side web application – no installation, no server, no data upload

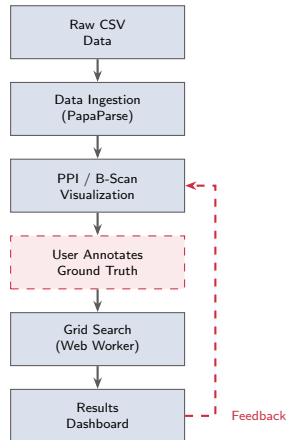
Technology stack:

- React 19 + TypeScript
- Vite build system
- Plotly.js for interactive visualization
- PapaParse for CSV ingestion
- Web Worker for grid search

Key features:

- PPI and B-scan visualization
- Multiple colormaps
- Interactive blob annotation
- Multi-frame temporal analysis
- Real-time progress updates

Processing pipeline:



Results: PPI View with Detections

figures/ppi_full_view_detections_and_ground_truth.png

Results: Full CFAR Studio Interface

figures/full_ui_ppi_with_optimization_results.png

Results: B-Scan View

figures/bscan_view_grayscale_with_cfar_detections.png

Results: Optimization Findings

Parameter space: Grid search over ~ 500 configurations in <10 seconds

Guard cells N_g :

- $N_g \geq 3$ improves scores for extended targets
- Prevents target self-contamination of noise estimate
- Point targets less sensitive to guard cell count

Reference cells N_r :

- Optimal range: $N_r \in [10, 20]$
- Balance: noise estimation accuracy vs. spatial homogeneity
- Large $N_r > 25$ degrades performance at terrain transitions

False alarm probability P_{FA} :

- Scene-dependent optimal value
- Open water: lower P_{FA} acceptable
- Near clutter edges: moderate P_{FA} maintains detection

Variant comparison:

- **GO-CFAR:** fewer false alarms, conservative
- **SO-CFAR:** higher detection, more false alarms
- **CA-CFAR:** best overall in homogeneous scenes

Temporal stability: Top configs achieve >0.9 across all frames

Discussion and Limitations

Strengths:

- Formalizes operator knowledge as quantitative objective
- Exhaustive search finds scene-specific optimal config
- Area-weighted scoring more informative than binary detection
- Temporal stability identifies consistent configs
- Zero-infrastructure deployment (browser-only)
- Open source, no licensing constraints

Applications:

- Autonomous vessel mapping and deconfliction

Limitations:

- Assumes quasi-stationary targets and fixed platform
- Moving targets require frame registration or tracking
- Browser constraints: no GPU, single-threaded JS
- Practical CSV file size limit: ~ 50 MB
- Grid search feasible for current 3-parameter space
- More variants (OS-CFAR) would benefit from Bayesian optimization

Future work:

- Extend to additional CFAR variants
- Support for moving target annotation

Questions?

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